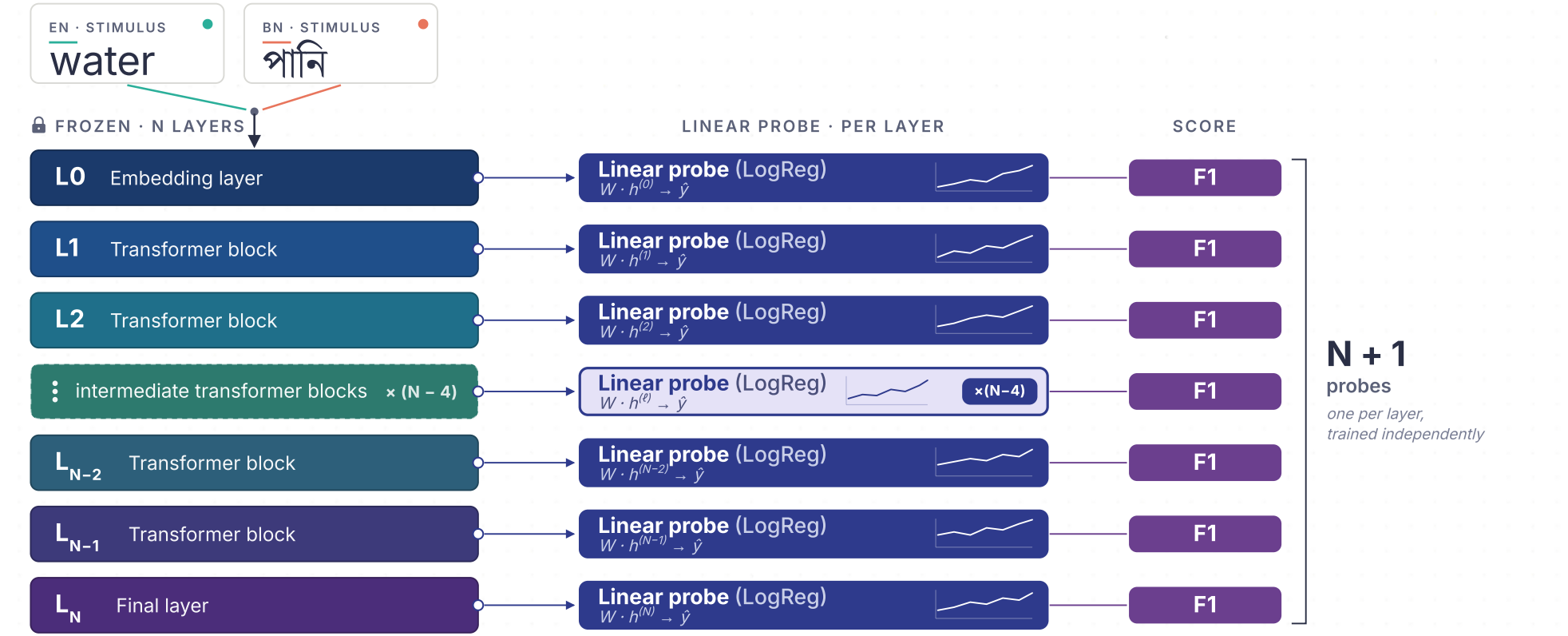
Per-layer linear probing of frozen multilingual LLMs

FIGURE · METHOD SCHEMATIC

One linear probe trained per hidden layer · evaluated in-language (EN → EN) and as zero-shot cross-lingual transfer (EN → BN)



EVALUATION · EACH PROBE TESTED TWICE

LANE A · IN-LANGUAGE METRIC

English-trained probe → Held-out English → In-language F1

$\hat{y} = W \cdot h^{(\ell)}$ · 5-fold CV · unseen at probe-train time · how well does ℓ encode the property at all?

METRIC macro-F1 ↑

LANE B · CROSS-LINGUAL TRANSFER METRIC

Same probe (no retraining) → Bangla zero-shot (পানি) → Transfer F1

trained on EN only · probe sees no Bangla during training · how cross-lingual is layer ℓ's geometry?

METRIC macro-F1 ↑